# Ott Pluvio$^2$ Maintenance

**Important:**

Before performing maintenance to the sensor, please check thoroughly the details of that procedure in this document, if any questions arise do not hesitate to contact David Trejo (trejo_cancino.david_ulises@uqam.ca).

After conducting any maintenance protocol, record it in the EcoFlux Inventory notes to have a follow-up of each sensor and station. If something is wrong with any equipment, please contact David Trejo Cancino (trejo_cancino.david_ulises@uqam.ca) to register it in the database and proceed with the required diagnostics and repairs.

## Summary

| Procedure | Frequency | Items required |
|---|---|---|
| Emptying the collecting bucket | When bucket is more than 80% full | - Protective gloves |
| Adding anti-freeze for winter operation | When temperatures are continuously below 0°C | - Anti-freeze<br>- Water<br>- Protective goggles |
| Performing visual inspection | As needed | |
| Additional checks in cases of defect | As needed | |
| Troubleshooting/remedy | As needed | |
| Performing the Guided Accuracy Test | As needed | - PC<br>- USB cable |

## Emptying the collecting bucket

**Caution:**

Be careful when emptying the collecting bucket, a completely filled bucket has a weight of up to 35 kg.

There is a risk of fingers being crushed when loosening/tightening the knurled screws and when fitting the pipe housing, wear protective gloves when emptying.

**When to empty?**

The bucket can be emptied at any time but must be emptied when the collecting bucket level is 80%. A bucket status warning will appear after the SDI-12 command. The warning **'+1'** will be the 9$^{th}$ value in the output from the command **"aD2!"** after **"aM!"**, **"aMC!"**, **"aC!"** or **"aCC!".** Typically, this needs to be done once or twice a year.

1. **Loosen three knurled screws.**
2. **Remove the pipe housing upwards**
3. **Remove collecting bucket carefully**
4. **Remove both spring-loaded terminals and carefully remove the collecting bucket.**
   Note: Proceeding carelessly (roughly placing the filled collecting bucket on the bucket support, may damage the weighing mechanism)
5. **Empty the collecting bucket**
6. **Place the collecting bucket on the bucket support.**
7. **Align the pipe housing and install it, refer to Fig. 1.**
8. **Retighten three knurled screws.**

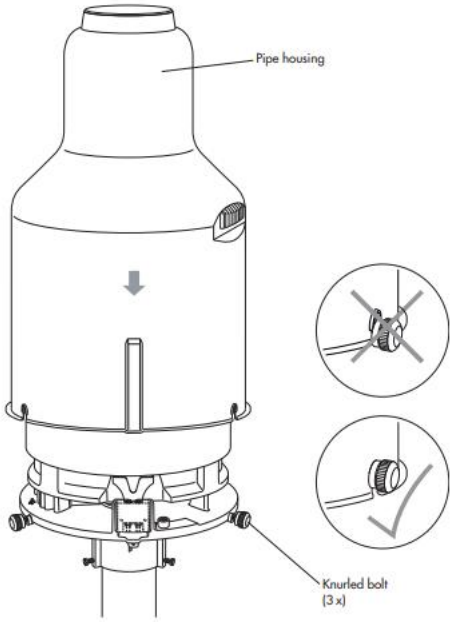



Figure 1: Fitting the pipe housing and tightening the knurled screws

## Adding Anti-freeze agent for winter operation

**Important:** Risk of eye injuries from anti-freeze agent, wear protective goggles when filling/emptying the anti-freeze agent.

**Note:** For temperatures below 0 °C, anti-freeze agent should be added to the collecting bucket. This causes the collected solid precipitation to gradually thaw in the collecting bucket. **Moreover, it prevents the unit from getting damaged when the collecting bucket is filled for more than 80% and the contents are frozen.** Use the anti-freeze POWERCOOL DC 924-PXL in an aqueous solution (manufacturer: Thermochema GmbH, A-4460 Losenstein; Phone +43 7255 4244-0; www.thermochema.at.).

1. **Preparing the anti-freeze solution: Mix 5 liters of anti-freeze with 2 litres of water.**
2. **Carefully add anti-freeze solution to the collecting bucket (the pipe housing does not have to be removed for this).**

**Note:** Add anti-freeze agent only as an aqueous solution (add 40% of water), never use neat! (POWERCOOL is hygroscopic; this can lead to the output of false precipitation.)

**Note on winter operation without anti-freeze:** The unit may be operated in the winter without adding anti-freeze solution. However, this mandatorily requires the collecting bucket to be checked and emptied in regular intervals.

**Note:** When the accumulated precipitation freezes and has filled the collecting bucket up to 80%, this may result in permanent damage to the weighing mechanism.

**Note on third-party anti-freeze solution products:**

Third-party products may be used provided the following requirements are met:

- Good solubility in water, low density (i.e. the anti-freeze agent does not settle below the water
- Low evaporation (do not use methanol).
- Little corrosive effect on aluminium and stainless steel.
- Low freezing point even with high collecting bucket level.
- Low hygroscopic properties (absorption of moisture from ambient air, which would effect the measurement results).
- Note chemical compatibility with collecting bucket (polyethylene).
- No gumming after use in open containers for several months.

## Performing the visual inspection

- Check smooth movement of the collecting bucket in all directions at the lower edge of the bucket orifice. To do this, with the pipe housing closed, move the bucket slightly with a sideways tap on the inside. The collecting bucket edge may not contact the pipe housing! A short movement can affect the measured value “Intensity RT” (short delay time)!
- If present, carefully remove any contamination (e.g. insects, insect nests, spider webs, etc.,), and icing.

## Additional checks in cases of defect

- Transportation lock removed?
- Is the collecting bucket distorted?
- Is the collecting bucket correctly positioned on the bucket support?
- Is the pipe housing correctly positioned and not damaged?

- Is the moving part of the weighing mechanism contacting other components, e.g. caused by contamination?
- Is the red LED flashing?

If in doubt, perform a Guided Accuracy Test.

# Troubleshooting/remedy

## General

- OTT Pluvio² L is connected to PC/tablet via USB cable (Pluvio unit disconnects communication over the SDI-12 interface); red LED is flashing once every two seconds, green LED permanently lit
  - **Disconnect USB cable.**
- Precipitation gauge supply voltage is missing, incorrectly connected, or is outside the specified range; red LED is continuously OFF:
  - **Restore power supply to the unit (replace fuse as necessary);**
  - **Check supply voltage level and adjust as necessary;**
  - **Check supply voltage wiring and correct as necessary.**
- Precipitation guage and the data recording unit connected are powered by two isolate power sources; GNDs of the two power supplies are not connected to each other:
  - **Establish connection between the two GNDs.**

## SDI-12 Interface

- Incorrect configuration of the OTT Pluvio² L and/or of the data recording unit connected:
  - **Check the SDI-12 sensor address of the OTT Pluvio² L and correct as necessary** (using the OTT Pluvio² L Operating Software)**;**
  - **Check the "Communication interface" (SDI-12) setting and correct as necessary** (Using the OTT Pluvio² L Operating Software)**;**
  - **Check the configuration of the data recording unit connected and correct as necessary** (refer to the data recording unit Operating Instructions)**.**

## RS-485 Interface – SDI-12 protocol

- RS-485 interface wiring is incorrect (pins "Data RS-485…..")
  - **2 – wire: check pin assignments of "… - A", "… - B" and swap as necessary;**
  - **4 – wire: check pin assignments of "… TX – A", "… TX – B" as well as of "… RX – A", "… RX – B" and swap as necessary.**
- Incorrect configurations of the OTT Pluvio² L and/or of the data recording unit connected:
  - **Check the SDI-12 sensor address of OTT Pluvio² L and correct as necessary** (using the OTT Pluvio² L);
  - **Check the "Communication interface" (RS-485/RS-485 4-wire) setting and correct as necessary** (using the OTT Pluvio² L Operating software);

  - **Check the "RS-485 protocol type" (SDI-12 protocol) setting and correct as necessary** (using the OTT Pluvio² L Operating Software);
  - **Check the configuration of the data recording unit connected and correct as necessary** (refer to the data recording unit Operating Instructions)**.**
- When an RS-485 interface converter is used – incorrect settings:
  - **Check settings and correct as necessary.**
- When terminal software (e.g. for testing) is used – incorrect settings:
  - **Check settings and correct as necessary.**

### Loss of communication through the USB interface

- Connection between OTT Pluvio² L and PC/tablet via USB cable failed; red LED is flashing once in two seconds; green LED is OFF:
  - **Check connection via USB cable and correct as necessary.**
- USD interface driver on PC/tablet is not or incorrectly installed:
  - **(re)install USB interface driver.**

### No/low precipitation output

- No precipitation output when USB interface is/was connected, after emptying, or during start-up/power failure:
  - **Wait 5 minutes.**
- Collecting bucket overflow:
  - **Empty the collecting bucket.**
- Evaporation rate is very high (collecting bucket is empty and ambient temperature is high):
  - **Add approx. 1 to 2 litres of water to the collecting bucket.**
- Apparently low output for NRT values:
  - **Wait for 5 minutes** (delay caused by filter algorithm)**.**
- Increase in weight > approximately 12 mm within 6 seconds:
  - **No precipitation output (filter algorithm detects foreign matter), since amount of natural precipitation is exceeded; check precipitation gauge status: response to SDI command** aD2!; <value 9> = +16 (Using the OTT Pluvio² Operating Software).
- Precipitation gauge probably failed:
  - **Perform Guided Accuracy Test** (using the OTT Pluvio² Operating Software).
- Precipitation gauge is not level:
  - **Check the bubble level indicator and correct adjustment as necessary.**

**Note:** Check whether measured value used is suited for present application (behaviour over time: Real-time output (RT), Non Real-time output (NRT), or combination value (RT-NRT))!

### Incorrect Precipitation Output

- Foreign matter (leaves, insects, …) have entered the collecting bucket (increase in weight < approx. 12 mm within 6 seconds; any higher increase is detected as foreign matter by the filter algorithm
  - **Manually correct the amount of precipitation by deducting a respective amount.**

- Base plate is incorrectly levelled:
  - **Rotate the base plate so that the knurled screw next to the bubble level indicator is facing south direction.**
- Anti-freeze agent without water was added (hygroscopic behaviour):
  - **Add anti-freeze agent only as an aqueous solution (add 40% more water)**
- … in the measured values “Intensity RT” + “Accu RT-NRT” at measuring sites exposed to wind:
  - **Alternately use measured value “Accu NRT”. (The measured values “Intensity RT” + “Accu NRT” are more susceptible to wind due to measurement technology.**
- … at the pulse output at measuring sites exposed to wind:
  - **If possible, use the SDI-12 interface and the measured value “Accu NRT”. (The pulse output is based on the measured value “Accu RT-NRT” which is more susceptible to wind due to measurement technology.)**

## Performing the Guided Accuracy Test

Using the OTT Pluvio$^2$ Operating Software, a Guided Accuracy Test may be performed. Using this test, the OTT Pluvio$^2$ L can be easily and quickly checked for proper operation on site.

In addition to this feature, other operations are provided by this Operating Software: e.g. modify OTT Pluvio$^2$ L basic settings, start measurements, or update the firmware.

**Note**: Perform the Accuracy Test on calm days without precipitation! (Otherwise, the test will be affected by wind and unwanted precipitation.) The USB interface also supplies operating voltage to the OTT Pluvio$^2$ L. No additional power supply is required. However, if present, it does not have to be removed.

**Preparatory work**

Prerequisite: PC running Microsoft Windows 7 or later.

- Log onto the PC as an administrator
- Installing the OTT Pluvio$^2$ Operating Software:
  - Copy the file “setup.exe” into a directory of the PC;
  - Start the file “setupe.exe” the “InstallShield Wizard” (installation program) opens;
  - Follow the installation instructions on the screen.
- Installing the USB interface drivers (FTDI drivers)
  - Copy the file “CDM v2.12.00 WHQL Certified” into a directory of the PC;
  - Remove the USB interface cover and connect the OTT Pluvio$^2$ L to the PC via the USB cable (refer Fig.2);
  - Complete the device driver installation wizard that is automatically started;
  - Start the file “CDM v2.12.00 WHQL Certified”.

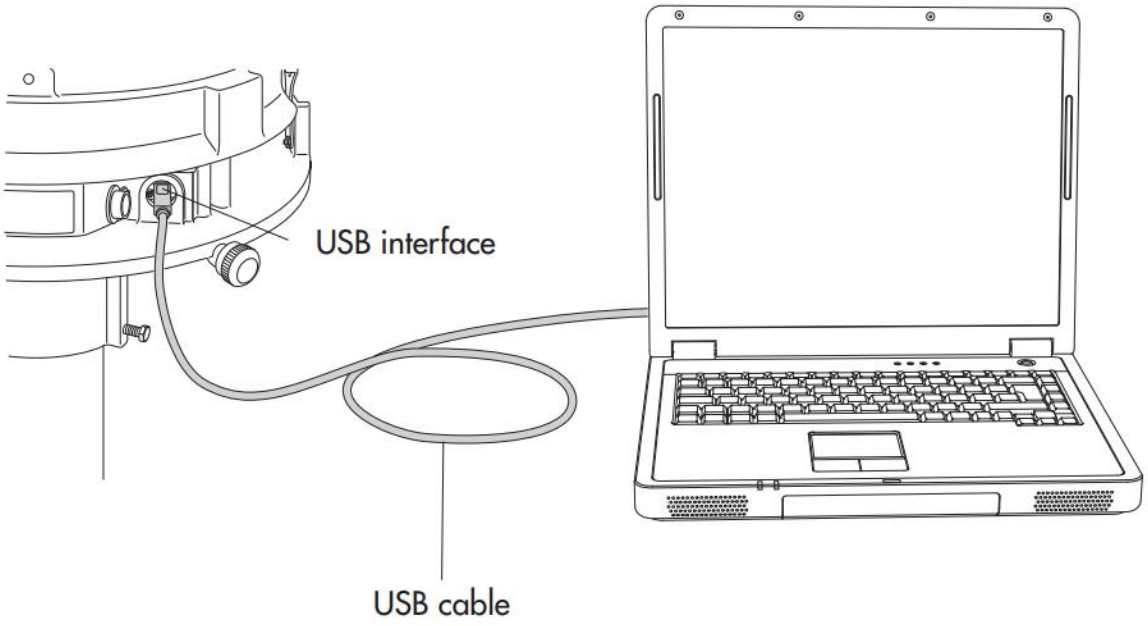



Figure 2. Connecting the OTT Pluvio$^2$ L to the PC using the USB interface. Once the PC supplies power to the OTT Pluvio$^2$ L via the USB interface, a green LD below the USB interface is lit.

## Performing the Guided Accuracy Test

- Loosen the three knurled screws at the pipe housing.
- Remove the pipe housing upwards.
- Remove collecting bucket carefully.
- Remove both spring-loaded terminals and carefully remove the collecting bucket.
- Remove the USB interface cover and connect the OTT Pluvio$^2$ L to the PC via the USB cable (Fig. 2).
- Start the OTT Pluvio$^2$ L Operating Software
- Select the “Connect” button → The Operating Software will establish a connection to the OTT Pluvio$^2$ L unit and retrieves different status information.
- Select the “Guided Accuracy Test” button → a wizard will guide you through the accuracy test. Make sure you follow all the steps and notes shown on the display!
- After completing the accuracy test, remove the USB cable.
- Replace the USB interface cover.
- Empty the collecting bucket as necessary and reinstall it.
- Align the pipe housing and install it.
- Retighten three knurled screws.

**Note:** After attaching the USB interface, the OTT Pluvio$^2$ L disconnects communication over the other interfaces.